

DR-AGENTOS: A PERSISTENT EMBODIED AGENT OPERATING SYSTEM WITH 4D WORLD MEMORY

Anonymous authors

Paper under double-blind review

ABSTRACT

Long-horizon embodied agents must remember a changing world, retrieve the right evidence for the current task, and prevent uncertain model outputs from silently becoming physical facts. We present DR-AGENTOS, a persistent embodied agent operating system that couples a typed agent controller with an evidence-grounded 4D World Memory. The memory lifts multimodal observations into a shared 3D frame, maintains stable entity identities with time-valid state versions, and links every assertion to its sensor provenance. A task-conditioned retrieval service combines descriptions, object crops, scene context, and explicit relations through rank fusion; a constrained vision-language model may select only a retrieved entity or abstain, while coordinates remain owned by mapping. Typed tools, closed-loop verification, conflict-aware reflection, and evidence-gated updates then connect memory to action without granting the language model authority over world state. We evaluate these memory-facing capabilities on LMEE-Bench and OpenEQA. The resulting design treats memory as an auditable system service rather than prompt context, enabling agents to query, act on, and responsibly revise experience over long horizons.

1 INTRODUCTION

Embodied agents increasingly translate open-ended instructions into actions grounded in perception. Foundation models broaden the goals robots can interpret, while open-vocabulary maps and scene representations provide semantic grounding beyond fixed object classes (Ahn et al., 2022; Driess et al., 2023; Huang et al., 2023; Gu et al., 2024). Real deployments, however, are not isolated episodes in static scenes. A robot may revisit a building over days or months, receive tasks that depend on earlier observations, and encounter changed objects, access conditions, or routes. It must therefore preserve and revise a grounded account of the world, not merely recognize the current view or predict a plausible next action.

Existing work addresses complementary pieces of this problem. Semantic maps and hierarchical scene graphs organize spatial entities and relations (Huang et al., 2023; Gu et al., 2024; Werby et al., 2024); dynamic and lifelong mapping models changes in geometry or object state (Bescos et al., 2018; Narayana et al., 2020; Liu et al., 2024); and embodied-agent frameworks connect foundation-model reasoning to callable skills and multimodal memory (Zhou et al., 2026; Tian et al., 2026; Wang et al., 2026a; Li et al., 2026). Yet task context, spatial memory, tool execution, and state update are often maintained by separate components without a shared contract for identity, temporal validity, provenance, and execution authority.

That missing contract creates a practical failure mode: a fluent controller can convert uncertain observations or successful software calls into unsupported facts. Object coordinates must originate from mapping, robot poses from the execution backend, and task completion from observed outcomes rather than a planner’s return value. Likewise, disagreement with memory may indicate a persistent change, but it may also arise from occlusion, association error, or a transient condition. Long-term operation thus needs both an evolving world model and an explicit policy governing which evidence may change it.

We present DR-AGENTOS, a persistent embodied agent operating system built around a typed Agent Controller and a 4D World Memory. The memory represents 3D semantic structure and temporal evolution under persistent identities. Observations are projected into a common frame and organized as a **Timeline–Floor–Room–Object** hierarchy; mutable attributes occupy time-valid versions, and every assertion remains linked to observations, poses, timestamps, and confidence. Here, “4D” denotes an executable, reconstructable world model rather than a sequence of timestamped frames.

Access to this memory is task conditioned. A structured query selects a causal memory view, after which multimodal retrieval combines descriptions, object crops, scene context, and relations. A constrained vision-language model (VLM) may select only from retrieved entities or abstain; it cannot invent an identity or edit a coordinate. The same source-grounded discipline governs execution: typed tool calls pass through validation, returned evidence is checked before task progress is committed, conflicts trigger active verification, and only corroborated changes become new persistent versions.

We evaluate memory-facing capabilities on LMEE-Bench (Wang et al., 2026b) and OpenEQA (Majumdar et al., 2024). These benchmarks test long-term exploration memory and episodic embodied question answering across different environments and question types.

Contributions. Our contributions are:

- an embodied AgentOS that separates high-level reasoning from authoritative tool execution, outcome verification, and state commit;
- an evidence-grounded 4D World Memory unifying 3D semantics, spatial hierarchy, persistent identity, temporal versions, and observation provenance; and
- task-conditioned retrieval with multimodal rank fusion, relation-aware queries, constrained candidate resolution, and evidence-gated state evolution.

2 RELATED WORK

Semantic and dynamic world models. Visual Language Maps attach language-aligned features to maps for open-vocabulary navigation (Huang et al., 2023), while ConceptGraphs and HOV-SG organize entities in open-vocabulary 3D scene graphs (Gu et al., 2024; Werby et al., 2024). These representations support spatial retrieval but principally describe a scene state. DynaSLAM removes dynamic content during mapping (Bescos et al., 2018); lifelong semantic mapping and DynaMem instead update maps as environments evolve (Narayana et al., 2020; Liu et al., 2024). Other work represents time-indexed 3D content or human-centered events (Gorlo et al., 2025; Bartoli et al., 2026). DR-AGENTOS builds on these lines by treating temporal validity, stable identity, and source evidence as a shared service used by reasoning and execution.

Embodied agents and memory. SayCan grounds language-model decisions in skill affordances, and PaLM-E injects embodied observations into a multimodal language model (Ahn et al., 2022; Driess et al., 2023). Recent systems increasingly combine foundation models, spatial memory, and tool use (Zhou et al., 2026; Tian et al., 2026; Wang et al., 2026a; Li et al., 2026; Xin et al., 2026). Long-term-memory and EQA benchmarks test whether stored experience supports later reasoning (Majumdar et al., 2024; Wang et al., 2026b). Our focus is the systems boundary between these components: the controller may interpret goals and choose tools, but mapping, execution, verification, and evidence-gated commit retain authority over physical facts.

3 DR-AGENTOS

3.1 PROBLEM SETTING AND SYSTEM LOOP

At decision round t , the agent receives the available subset of RGB image $\mathbf{I}_t$, depth $\mathbf{D}_t$, geometric observation $\mathbf{L}_t$, robot pose ξ_t , and timestamp τ_t . Given natural-language task $\mathcal{T}$, execution state s_t , active task stage ρ_t , callable tools $\mathcal{Z}$, and persistent memory $\mathcal{M}_t$, the controller constructs a

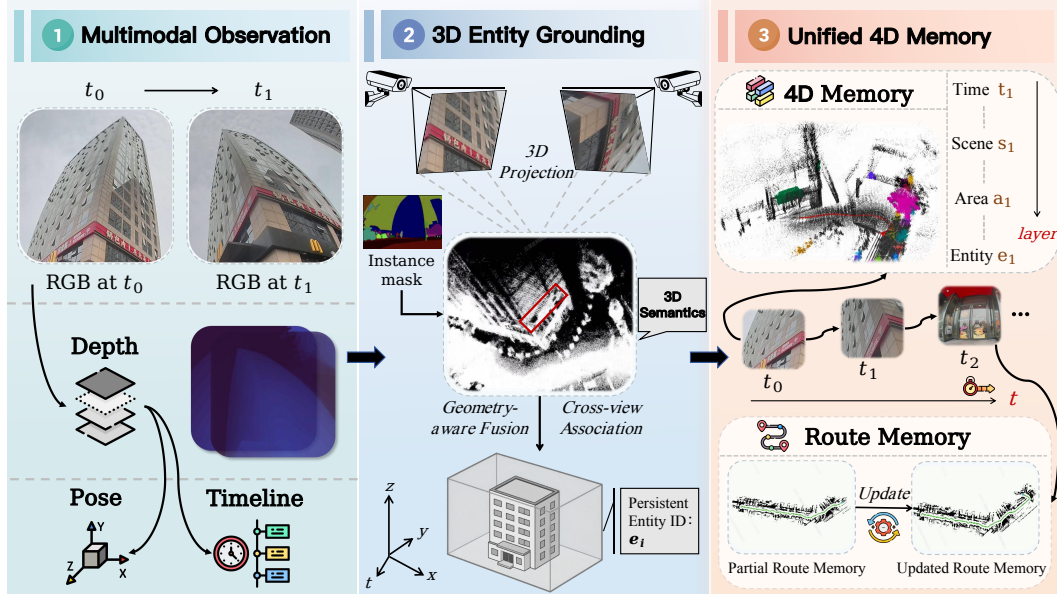


Figure 1: Construction of 4D World Memory. Multimodal observations are registered in a shared world frame, consolidated under persistent entity identities, and stored as time-valid world and route memory.

task-conditioned query and selects a tool call:

$$\mathbf{q}_t = \mathcal{Q}(\mathcal{T}, \rho_t, \mathbf{s}_t), \quad (1)$$

$$(z_t, \mathbf{a}_t) = \Pi(\mathcal{T}, \rho_t, \mathbf{s}_t, \mathcal{R}(\mathcal{M}_t, \mathbf{q}_t), \mathcal{Z}). \quad (2)$$

The controller then verifies execution against explicit completion conditions, detects conflicts with existing memory, gathers evidence when needed, and applies a gated update. This yields a closed perception–memory–reasoning–action–reflection loop rather than a one-shot perception-to-action pipeline.

3.2 EVIDENCE-GROUNDED 4D WORLD MEMORY

Direct semantic projection. For pixel $\tilde{\mathbf{u}} = [u, v, 1]^\top$ with valid depth, the corresponding world point is

$$\mathbf{x}^w = \text{proj}_3 \left(\mathbf{T}_{w \leftarrow c, t} \begin{bmatrix} D_t(u, v) \mathbf{K}^{-1} \tilde{\mathbf{u}} \\ 1 \end{bmatrix} \right), \quad (3)$$

where $\mathbf{K}$ is the camera intrinsic matrix and $\mathbf{T}_{w \leftarrow c, t}$ the camera-to-world transform. Semantic regions from open-vocabulary perception are lifted directly into this frame. Cross-view consolidation uses semantic compatibility, 3D proximity, temporal consistency, and source confidence, so repeated observations strengthen a persistent entity rather than create disconnected detections.

Causal, versioned memory. Memory at time t is materialized only from past and current evidence,

$$\mathcal{M}_{\leq t} = \text{Materialize}(\Delta \mathcal{M}_0, \dots, \Delta \mathcal{M}_t) = (\mathcal{G}_{\leq t}, \mathbb{T}_{\leq t}, \mathcal{B}_{\leq t}), \quad (4)$$

where $\mathcal{G}$ is a semantic-spatial graph, $\mathbb{T}$ a set of world-state segments, and $\mathcal{B}$ an evidence/provenance bank. An entity has a stable identity i , semantic label ℓ_i , geometry $\mathbf{p}_i$, relations $\mathcal{N}_i$, and a pointer to its current version. Its k -th state version is

$$h_i^k = ([\tau_{i,s}^k, \tau_{i,e}^k], y_i^k, c_i^k, \mathbf{f}_i^k, \mathcal{P}_i^k), \quad (5)$$

containing a validity interval, structured state, confidence, visual features, and provenance. Stable identity is therefore separated from mutable state: a door remains the same entity when its accessibility changes, and an object may move without losing its history. The graph follows a Timeline–Floor–Room–Object hierarchy and retains metric, containment, proximity, and connectivity relations. Assertions link back to sensor time, source pose, visual support, and confidence.

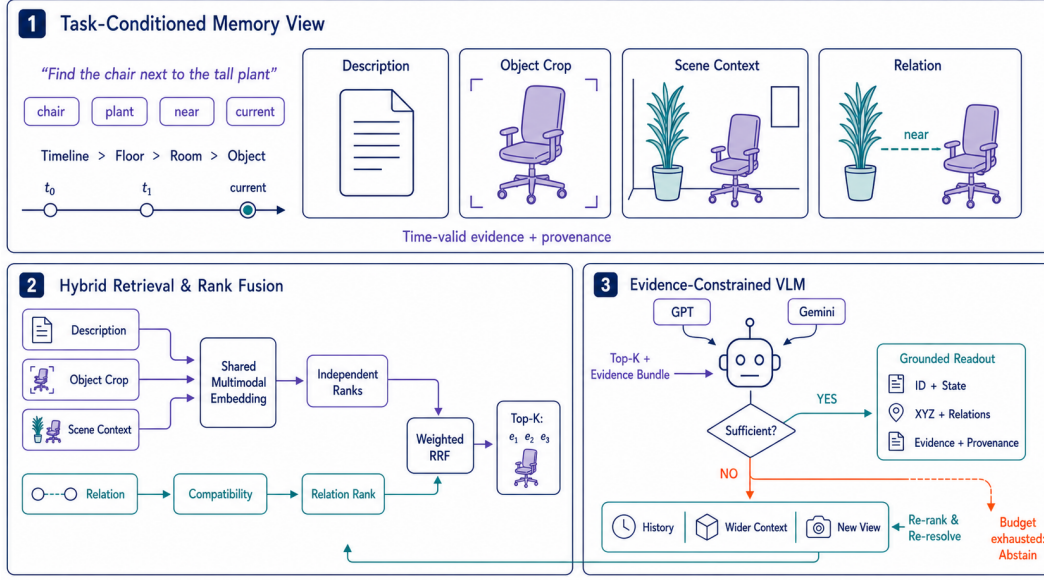


Figure 2: Task-conditioned retrieval. A causal memory view supplies semantic, visual, contextual, and relational evidence; rank fusion generates candidates; a constrained VLM confirms a candidate, requests more evidence, or abstains.

3.3 TASK-CONDITIONED MULTIMODAL RETRIEVAL

Each entity’s retrieval record contains its current description, object crop, full-frame context, mapper-owned coordinate, source pose, and provenance. The query is encoded against three semantic channels—description, crop, and context—and ranked independently. For normalized query embedding $\mathbf{z}_q$ and entity representation $\mathbf{z}_i^{(m)}$,

$$s_i^{(m)} = \langle \mathbf{z}_q, \mathbf{z}_i^{(m)} \rangle, \quad m \in \{\text{desc}, \text{crop}, \text{ctx}\}. \quad (6)$$

We combine ranks rather than raw scores because channel scales need not be calibrated. Weighted Reciprocal Rank Fusion gives

$$S_i^{\text{RRF}} = \sum_{m \in \mathcal{M}_q} \frac{w_m}{K_{\text{RRF}} + r_i^{(m)}}, \quad (7)$$

with $K_{\text{RRF}} = 60$ and default weights $(1.0, 1.0, 0.5)$ for description, crop, and context.

Relation-aware QueryGraph. Relational instructions are parsed into a target, an anchor, and a supported spatial predicate. For a `near` relation, target i and anchor a receive

$$S_{i,a}^{\text{spatial}} = \alpha / r_a^{\text{sem}} + (1 - \alpha) / (1 + \|\mathbf{p}_i^{xy} - \mathbf{p}_a^{xy}\|_2), \quad (8)$$

where $\alpha = 0.45$. The best compatible anchor supplies a relation rank that enters Equation (7). Unsupported parses fail closed to the three-channel retrieval path.

Constrained VLM resolution. The top- K candidates are presented with their evidence. The VLM must return `confirmed`, `ambiguous`, or `not_found`; a confirmed identifier must belong to the presented set $\mathcal{C}_q$, otherwise it is rejected. Crucially, confirmation binds the semantic goal to the coordinate already stored by memory,

$$\hat{i} \in \mathcal{C}_q \text{ or } \hat{i} = \emptyset, \quad \mathbf{g}_{\text{sem}} = \mathbf{p}_{\hat{i}}. \quad (9)$$

The VLM selects among evidence-backed identities but cannot invent an entity or modify geometry. If evidence is insufficient, the controller can retrieve history, broaden context, acquire a new view, and re-rank within a fixed budget.

3.4 TYPED TOOLS AND CLOSED-LOOP EXECUTION

Natural language is normalized into goal G , constraints Ω , completion predicate χ , and a subtask dependency graph. Each tool declares a name, typed input schema, permitted controller modes, and schema version; each result returns execution status, structured payload, evidence handles, latency, and a normalized error. The registry validates tool existence, mode, arguments, and call budget before dispatch. This interface separates probabilistic reasoning from authoritative components: mapping owns entity identities and coordinates, workers own deterministic algorithmic outputs, the runtime owns pose and collision feedback, and verification owns state commit.

For navigation, an object coordinate may lie in occupied space, so the planner snaps it to the nearest cell in the robot’s reachable free component $\mathcal{R}_t$:

$$g' = \arg \min_{v \in \mathcal{R}_t} \|v - g\|_2^2. \quad (10)$$

An 8-connected weighted distance field selects a short-term goal with minimum remaining geodesic distance. Only one action is issued before updated pose and collision evidence trigger replanning. Thus a successful planning call is distinct from a verified physical outcome.

3.5 CONFLICT-AWARE REFLECTION AND UPDATE

New evidence is compared with current memory along state, position, visibility-aware absence, and topology dimensions. A missing detection counts as evidence of disappearance only if the remembered location should be visible given pose, field of view, occlusion, and sensing conditions. Material conflicts cause the controller to retrieve history, change viewpoint, repeat perception, inspect a route, or seek clarification.

For proposed state $\hat{h}_{i,t}$, let evidence item b have reliability r_b , agreement a_b , and source-dependent weight w_b . Corroboration is

$$C_{\text{evidence}} = \frac{\sum_{b \in B_t(i)} w_b r_b a_b}{\sum_{b \in B_t(i)} w_b + \epsilon}. \quad (11)$$

A persistent update is committed only when the conflict has been examined and the score exceeds an entity-dependent threshold:

$$g_t(i) = \mathbb{I}[C_{\text{evidence}} \geq \eta_i \wedge r_t^{\text{conf}}(i) = 1]. \quad (12)$$

When the gate is closed, evidence remains provisional; when open, the new state is appended as a version and the current pointer advances. Earlier versions are never overwritten. This design preserves uncertain observations for later reasoning without promoting them prematurely to durable facts.

4 EXPERIMENTAL SETUP

Benchmarks. We evaluate memory-facing behavior on **LMEE-Bench** (Wang et al., 2026b), which tests long-term memory over exploration histories, and **OpenEQA** (Majumdar et al., 2024), which tests embodied question answering over ScanNet and HM3D environments. For OpenEQA we use episodic-memory EQA: the agent answers from a fixed, fully observed episode rather than actively exploring.

Metrics and protocol. For LMEE-Bench we follow the official multiple-choice and open-answer protocols, reporting choice accuracy and normalized open-answer score on Easy, Medium, Hard, and overall splits. The categories cover attributes, state, location, relations, and counting. For OpenEQA we follow environment-based reporting and use LLM-Match on ScanNet, HM3D, and their combined question set. Ground-truth answers are withheld from the agent and used only for scoring. Frame budgets are reported separately because the benchmarks sample episode history differently.

Table 1: LMEE-Bench results. Choice accuracy and open-answer score are reported by difficulty. Dashes denote experiments that remain to be run; this table is not submission-ready until they are resolved.

Method	Memory/Input	Frames	Choice accuracy (%) $\uparrow$				Open-answer score $\uparrow$			
			E	M	H	All	E	M	H	All
Direct VQA	Sampled frames	–	–	–	–	–	–	–	–	–
MemoryExplorer (Wang et al., 2026b)	Learned memory	–	–	–	–	–	–	–	–	–
ConceptGraphs (Gu et al., 2024)	3D scene graph	–	–	–	–	–	–	–	–	–
DR-AGENTOS (ours)	4D world memory	–	–	–	–	–	–	–	–	–

Table 2: OpenEQA episodic-memory results. LLM-Match is split by environment. Dashes denote pending experiments, not zero-valued results.

Method	Memory/Input	Frames	ScanNet	HM3D	All
Direct VQA	Sampled frames	–	–	–	–
Caption memory	Text history	–	–	–	–
ConceptGraphs (Gu et al., 2024)	3D scene graph	–	–	–	–
DR-AGENTOS (ours)	4D world memory	–	–	–	–

Comparisons. We distinguish direct multi-frame VQA systems, which answer from sampled frames without persistent scene state, from memory-based systems using captions, scene graphs, snapshots, or learned long-term memory. DR-AGENTOS uses the same persistent 4D memory and task-conditioned retrieval interface in both evaluations. Exact model versions, prompts, inference settings, hardware, and run-level aggregation must be recorded with the final benchmark results; the required reporting checklist appears in Appendix A.3.

5 RESULTS

The repository does not yet contain completed benchmark outputs, so we retain explicit placeholders instead of presenting unverified values. Before submission, every entry in Tables 1 and 2 must be replaced by a measured value or a cited published result, and the surrounding analysis must report uncertainty and failure cases.

Required analyses. The final empirical section should test more than aggregate QA quality. At minimum, it should ablate temporal versions, provenance-aware retrieval, relation rank, constrained VLM resolution, and evidence-gated updates; report retrieval recall before and after VLM verification; measure latency and memory growth; and stratify errors into retrieval failure, stale-state selection, relation failure, hallucinated resolution, and scoring mismatch. These tests directly correspond to the paper’s system claims.

6 CONCLUSION

We introduced DR-AGENTOS, a persistent embodied agent operating system that treats memory as an evidence-grounded service shared by perception, reasoning, and action. Its 4D World Memory preserves identity, temporal validity, geometry, and provenance; task-conditioned retrieval exposes a bounded evidence set; and typed tools, constrained resolution, reflection, and gated updates prevent a language model from silently acquiring authority over world state. This architecture provides a concrete basis for auditing how embodied agents remember, act on, and revise experience. Completing the benchmark and ablation matrix in Section 5 is necessary before drawing comparative empirical conclusions.

7 LIMITATIONS

The current formulation assumes sufficiently accurate pose and depth for cross-view association; accumulated localization error can split identities or corrupt spatial relations. Rule-based relation parsing and evidence thresholds favor audibility but cover fewer expressions than unconstrained generation. A bounded VLM can still choose the wrong legal candidate, while active verification increases latency and does not guarantee disambiguation. Memory retention also creates privacy, storage, and security risks in real deployments. Finally, the current repository lacks completed benchmark and ablation results, so no quantitative superiority claim is warranted from this draft.

ETHICS STATEMENT

DR-AGENTOS is intended for research on embodied navigation, memory, and question answering. Persistent visual memory may capture people or private environments, and an erroneous retrieved state may cause unsafe physical behavior. Deployments should therefore obtain appropriate data consent, enforce retention and access controls, retain provenance for audit, and place platform-specific safety monitors outside the language-model controller. The benchmarks used here contain no new human-subject study by the authors.

REPRODUCIBILITY STATEMENT

Section 3 specifies the memory records, retrieval and fusion rules, constrained resolver, typed tool interface, navigation rule, and update gate. Section 4 defines the benchmark protocols and metrics; additional implementation and evaluation details are provided in the appendix. An anonymized code snapshot should accompany the review submission.

AI USE STATEMENT

Generative AI tools were used to edit and restructure the manuscript, improve readability, check LaTeX formatting, and assist with local engineering. They were not used to generate benchmark observations or reported evaluation scores. All AI-assisted text and code were reviewed by the authors, who take responsibility for the final content, claims, and artifacts.

REFERENCES

- Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, et al. Do as i can, not as i say: Grounding language in robotic affordances. *arXiv preprint arXiv:2204.01691*, 2022.
- Ermanno Bartoli, Buwei He, Dennis Rotondi, Sebastian Koch, Federico Tombari, Kai O. Arras, Patric Jensfelt, Yixi Cai, and Iolanda Leite. Gesto: Human-centric spatio-temporal memory for reasoning in dynamic scenes. *arXiv preprint arXiv:2608.10886*, 2026.
- Berta Bescos, José M. Fácil, Javier Civera, and José Neira. Dynaslam: Tracking, mapping and inpainting in dynamic scenes. *IEEE Robotics and Automation Letters*, 3(4):4076–4083, 2018.
- Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, et al. Palm-e: An embodied multi-modal language model. *arXiv preprint arXiv:2303.03378*, 2023.
- Nicolas Gorlo, Lukas Schmid, and Luca Carlone. Describe anything, anywhere, at any moment. *arXiv preprint arXiv:2512.00565*, 2025.
- Qiao Gu, Alihusein Kuwajerwala, Sacha Morin, Krishna Murthy Jatavallabhula, Bipasha Sen, Aditya Agarwal, Corban Rivera, William Paul, Kirsty Ellis, Rama Chellappa, et al. Concept-graphs: Open-vocabulary 3d scene graphs for perception and planning. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 5021–5028. IEEE, 2024.
- Chenguang Huang, Oier Mees, Andy Zeng, and Wolfram Burgard. Visual language maps for robot navigation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10608–10615, 2023.

- Yijian Li, Changze Li, Hantian Shi, Jiaying Luo, Jiyuan Cai, Ming Yang, and Tong Qin. Agenticnav: Zero-shot vision-and-language navigation as a tool-calling harness, 2026. URL <https://arxiv.org/abs/2606.10577>.
- Peiqi Liu, Zhanqiu Guo, Mohit Warke, Soumith Chintala, Chris Paxton, Nur Muhammad Mahi Shafiullah, and Lerrel Pinto. Dynamem: Online dynamic spatio-semantic memory for open world mobile manipulation. *arXiv preprint arXiv:2411.04999*, 2024.
- Arjun Majumdar et al. OpenEQA: Embodied question answering in the era of foundation models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- Manjunath Narayana, Andreas Kolling, Lucio Nardelli, and Phil Fong. Lifelong update of semantic maps in dynamic environments. *arXiv preprint arXiv:2010.08846*, 2020.
- Jiayi Tian, Shiao Liu, Yuting Xu, Jia Lu, Zihao Guan, Honglin Han, Di Yang, Minqi Gu, Yifei Qian, Tianlin Zhang, et al. Abot-agentos: A general robotic agent os with lifelong multi-modal memory. *arXiv preprint arXiv:2607.10350*, 2026.
- Qi Wang, Tianyi Wang, Chengyang Li, Shikun Ban, Yurun Chen, Yizhong Ge, Jason Qin, Chengtai Li, and Wentao Zhu. Towards the harness of embodied agents, 2026a. URL <https://arxiv.org/abs/2608.11246>.
- Sen Wang, Bangwei Liu, Zhenkun Gao, Lizhuang Ma, Xuhong Wang, Yuan Xie, and Xin Tan. Explore with long-term memory: A benchmark and multimodal llm-based reinforcement learning framework for embodied exploration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026b.
- Abdelrhman Werby, Chenguang Huang, Martin Büchner, Abhinav Valada, and Wolfram Burgard. Hierarchical open-vocabulary 3d scene graphs for language-grounded robot navigation. *Robotics: Science and Systems*, 2024.
- Zihao Xin, Wentong Li, Yixuan Jiang, Ziyuan Huang, Bin Wang, Piji Li, Jianke Zhu, Jie Qin, and Shengjun Huang. Agentvln: Towards agentic vision-and-language navigation, 2026. URL <https://arxiv.org/abs/2603.17670>.
- Xiaolin Zhou, Liu Liu, Tingyang Xiao, Wei Feng, Fa Fu, Xinrui Meng, Xinjie Wang, Jialiang Han, Boyang Yu, Yun Du, Wei Sui, and Zhizhong Su. Holoagent-0: A unified embodied agent framework with 3d spatial memory, 2026. URL <https://arxiv.org/abs/2606.23565>.

A SUPPLEMENTARY MATERIAL

A.1 MEMORY AND RETRIEVAL RECORDS

The persistent graph stores stable entity identifiers separately from time-varying versions. Each entity version includes its validity interval, structured attributes, confidence, feature handles, and provenance. Provenance identifies the observation time, source pose, sensor or rendered view, visual support, and content identity. The controller accesses these records through a typed memory query rather than receiving an unbounded transcript of observations.

The default retrieval service embeds descriptions, object-centric crops without rendered boxes, and full-frame context in one multimodal embedding space. Bounding-box overlays are retained only as VLM- or human-readable verification evidence; poses and world coordinates remain structured metadata and are never encoded as model-generated geometry. Near-identical similarities are tie-aware to prevent file order from producing arbitrary rank differences. The current resolver receives at most five candidates.

A.2 TYPED TOOL CONTRACT

A tool specification contains $(n, \sigma_{\text{in}}, \mathcal{C}, \nu)$: name, structured input schema, allowed controller modes, and schema version. A call contains a unique identifier, tool name, typed arguments, and version. A result contains execution status, structured payload, evidence references, latency, and a normalized error when execution fails. Representative services cover spatiotemporal memory query, exploration-state query, semantic object retrieval, map access, route planning, navigation, inspection, and change query. The registry rejects unknown names, invalid modes, malformed arguments, and calls beyond the current budget before invoking a worker.

A.3 EVALUATION REPORTING CHECKLIST

To make the completed study reproducible, the submission should report:

- exact benchmark releases, split sizes, frame sampling, and preprocessing;
- exact VLM and embedding-model versions, prompts, decoding parameters, and access dates;
- all retrieval weights, thresholds, candidate budgets, and tie tolerances;
- hardware, wall-clock latency protocol, repeated-run count, aggregation, and uncertainty;
- baseline sources and whether values are reproduced or quoted from prior work; and
- per-category failures and complete ablations aligned with each claimed component.

A.4 SUBMISSION COMPLIANCE NOTES

This source is configured for anonymous ICLR 2027 submission: \iclrfinalcopy remains disabled, author identities and affiliations are absent, references precede the appendix, and the required AI-use statement appears before the references. The initial-submission main text must end within page 9; references and this appendix are excluded from that limit. Authors should scan figure metadata, linked repositories, acknowledgments, PDF properties, and supplementary files for identifying information before uploading. The public project URL should not be included in the anonymous manuscript unless replaced with an anonymous artifact.